

Data for Slayer 150 1.01: what we use, what we rejected, and how much

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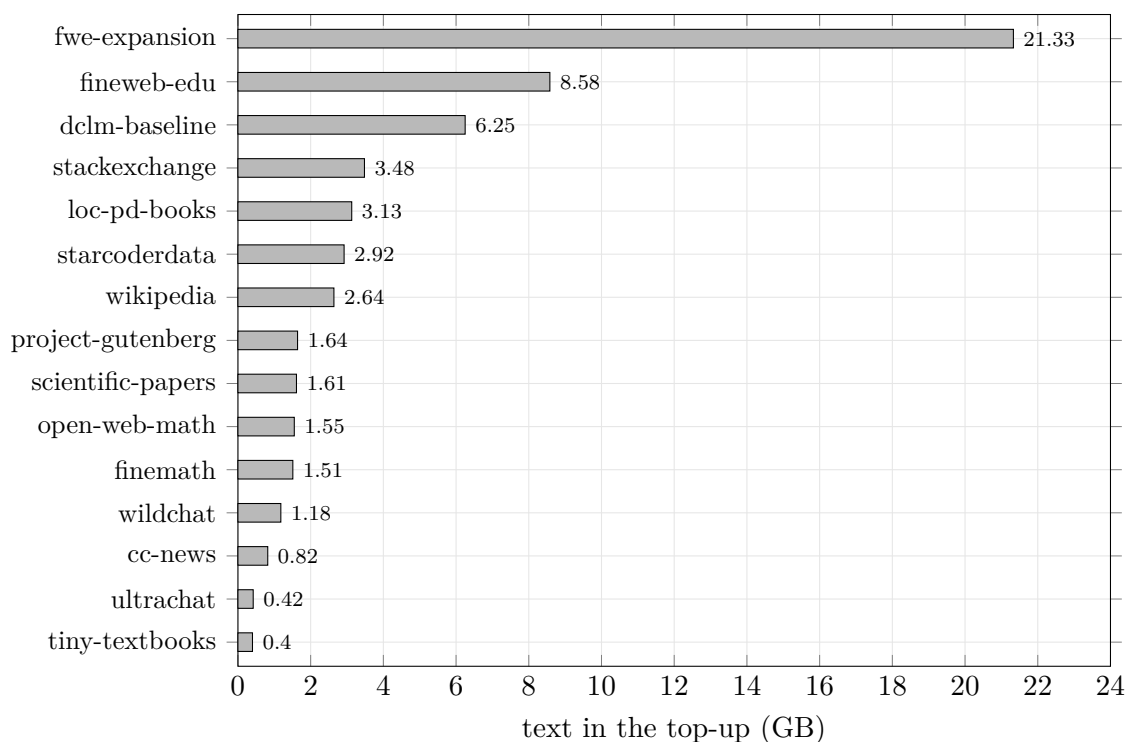


Figure 1: The stage-2 top-up by source (GB of UTF-8 text, 57.45 GB in total, 11,438,146 documents). Fifteen public sources at pinned revisions, kept in the source proportions of the base corpus scaled $\times 1.90$. Packing adds 8.46 GB of deliberate copies of ARC-similar *fwe-expansion* documents on top of these 57.45 GB. Source: build manifest `b73f45f5`.

1 Summary

Slayer 150 1.01 (139,279,760 parameters, 17 layers $\times$ 768, a 24,576-token BPE vocabulary) trains on about 24.7 billion tokens in two stages (Table 1). Every training document is seen once, with two deliberate exceptions described in Section 3: copies of ARC-similar documents and OpenStax repeated four times. There are no extra epochs: an ablation that repeated the pool lost (Section 6). The decay phase (the last 20 % of steps) uses the same mixture as the stable phase, because an ablation with an educational-text-heavy decay also lost.

Stage	Data	Tokens (24k)	Steps
1	ARC-MIX pool (the GoLLeM-v5 corpus), retokenized to 24k; first 6 of its 8 shards	6,455,623,680	0 $\rightarrow$ 197,010
2	one seeded document shuffle of the unused pool part and the top-up (Section 4.3)	18,266,295,978	197,010 $\rightarrow$ 754,453

Table 1: Training data stages. Batch $32 \times 1,024$ tokens per step.

2 Where to find the data

What	Location	Pin
ARC-MIX pool, 12k tokens	HF dataset <code>SlayerLab/gollem-ui-mirror</code> , file <code>arcmix/train.bin</code>	revision <code>486a3af3</code> , SHA <code>e87f594f</code>
ARC-MIX recipe card	HF <code>SlayerLab/gollem-v5-arcmix-9b</code> (card only)	—
12k tokenizer	HF <code>SlayerLab/gollem-v5-ckpts</code> , file <code>tokenizer.json</code>	SHA <code>37333075</code>
v5 validation stream	HF <code>SlayerLab/gollem-v5-ckpts</code> , file <code>vals/arcmix_val.bin</code>	SHA <code>8253a108</code>
Top-up sources	15 HF / S3 sources (Table 4)	pinned revisions
Base corpus (proportions)	HF <code>SlayerLab/minimal-en-corpus-5b</code>	revision <code>bb3aec23</code>

Table 2: Public locations. No step needs an HF token except two gated top-up sources (starcoderdata, tiny-textbooks); the token was used once, only inside one process environment, and deleted.

3 Stage 1: the ARC-MIX pool

The pool was built for GoLLeM-v5 with a 12k tokenizer and is reused unchanged in content (Table 3). Token shares of the final pool: base corpus 57.5 %, FineWeb-Edu expansion 29.3 %, ARC-similar copies 12.0 %, OpenStax 1.5 %. The component counts are taken before the removals below, so the shares sum to 100.3 %.

Component	Source	Tokens (12k)	Notes
Base corpus	<code>SlayerLab/minimal-en-corpus-5b@bb3aec23</code> (15 sources)	5,396,605,407	3,968,467 documents
FineWeb-Edu expansion	<code>HuggingFaceFW/fineweb-edu</code> , sample-100BT, first 2,384,000 rows	2,756,248,149	3,184 removed by decontamination
OpenStax	58 books, CC BY 4.0	140,742,204	$35,185,551 \times 4$
ARC-similar copies	FineWeb-Edu documents scored by <code>arc_score</code>	1,123,440,072	score ≥ 4 : +2 copies; score = 3: +1 copy

Table 3: ARC-MIX composition.

After removals the pool has **7,088,917 documents and 9,391,706,576 tokens** (12k, SHA `ecfd0a40`). We removed 5,537 documents: 4,362 by the WikiText-2 and ARC test/validation scan and 1,175 by the CC BY-NC-SA licence rule. Retokenized to 24k (decode with 12k, encode with 24k, round trip checked) it has **8,615,887,241 tokens** (SHA `8911b9c5`), built twice on two machines with the same SHA. One 24k token carries 9.0 % more text than one 12k token on this pool.

4 Stage 2: the top-up

4.1 Rule

The goal is about 16.4 billion further 24k tokens without repeating anything. We keep the **source proportions of the base corpus** (`minimal-en-corpus-5b`) scaled $\times 1.90$, the same FineWeb-Edu expansion and the same dose of ARC-similar copies as ARC-MIX (0.397 versus 0.398). A source that is short keeps what exists, and its missing bytes are spread over the other sources in proportion to their quotas. OpenStax is not scaled.

Build filters, in order: overlap with either held-out control set (Section 5.2) $\rightarrow$ duplicate of the base corpus $\rightarrow$ duplicate of FineWeb-Edu files already used $\rightarrow$ duplicate within the new data. Documents under 200 characters are dropped. A document is hashed as SHA-256 of its first and last 2,000 characters. The FineWeb-Edu file used as a control set (`000_00009`) is refused by its SHA. Build output: **11,438,146 documents, 57,450,444,352 bytes of text** (manifest SHA `b73f45f5`).

4.2 Sources

Source	Repository @ revision	Docs	GB	Share	Licence
fwe-expansion	HuggingFaceFW/fineweb-edu@87f09149	4,494,091	21.33	37.1 %	ODC-BY 1.0
fineweb-edu	same files, base-proportion part	1,809,246	8.58	14.9 %	ODC-BY 1.0
dclm-baseline	mlfoundations/dclm-baseline-1.0-parquet@817d6752	1,094,166	6.25	10.9 %	CC BY 4.0
stackexchange	togethercomputer/RedPajama-Data-1T@398f9257	1,468,456	3.48	6.1 %	CC BY-SA (content)
loc-pd-books	storytracer/LoC-PD-Books@5f1f4ee0	8,491	3.13	5.4 %	CC0
starcoderdata	bigcode/starcoderdata@9fc30b57	637,314	2.92	5.1 %	The Stack terms (gated)
wikipedia	wikimedia/wikipedia@b04c8d1c (20231101.en)	754,440	2.64	4.6 %	CC BY-SA 3.0 / GFDL
project-gutenberg	manu/project_gutenberg@164853d2	4,063	1.64	2.9 %	US public domain
scientific-papers	PMC Open Access, commercial-use subset	31,493	1.61	2.8 %	CC0 / CC BY / CC BY-SA
open-web-math	open-web-math/open-web-math@fde8ef8d	194,991	1.55	2.7 %	ODC-BY (upstream)
finemath	HuggingFaceTB/finemath@e92b25a6	263,340	1.51	2.6 %	ODC-BY
wildchat	allenai/WildChat-1M@7d6490e4	162,481	1.18	2.0 %	ODC-BY
cc-news	vblagoje/cc_news@81eb2ce0	313,380	0.82	1.4 %	unknown (named)
ultrachat	HuggingFaceH4/ultrachat_200k@8049631c	71,990	0.42	0.7 %	MIT
tiny-textbooks	nampdn-ai/tiny-textbooks@1f88d0fe	130,204	0.40	0.7 %	Apache 2.0 (gated)

Table 4: Top-up sources. Packing adds 1,484,830 copies of ARC-similar *fwe-expansion* documents (8.46 GB) on top of these files. cc-news and ultrachat are short sources; OpenSubtitles was dropped (Section 6).

4.3 The stage-2 region

The data after the switch checkpoint is one permutation with seed 20261007 of all documents of the unused pool shards and all top-up documents with their ARC copies, each followed by an end-of-text token. With the switch at step 197,010 the unused pool part is 1,777,155 documents and 2,153,971,445 tokens. The packed region has **18,266,295,978 tokens** (SHA 8bbd2c6a), 1.5 % fewer than the 762,940 steps of the original plan needed, so the run was shortened to 754,453 steps (decay from step 603,562) instead of repeating data. The top-up averages about 4.09 bytes per token, from 3.00 for code to 4.62 for tiny-textbooks, with FineWeb-Edu at 4.48 (Table 5). The region holds 47 more end-of-text tokens than units, because a few documents contain the end-of-text marker literally in their text; this does not change training, but unit boundaries must be taken from the packing index, not from the end-of-text tokens.

Source	Units	of which ARC copies	Tokens (24k)	Share
fwe-expansion	5,962,385	1,480,815	6,647,665,150	36.4 %
pool (unused shards)	1,777,155	—	2,153,971,445	11.8 %
fineweb-edu	1,804,284	—	1,916,743,396	10.5 %
dclm-baseline	1,088,455	—	1,499,403,178	8.2 %
stackexchange	1,468,446	—	1,036,377,059	5.7 %
starcoderdata	637,310	—	974,144,383	5.3 %
loc-pd-books	8,477	—	933,188,923	5.1 %
wikipedia	752,475	—	664,931,978	3.6 %
finemath	262,926	—	455,661,822	2.5 %
scientific-papers	31,491	—	448,862,907	2.5 %
project-gutenberg	4,050	—	434,101,961	2.4 %
open-web-math	194,863	—	430,112,118	2.4 %
wildchat	162,462	—	297,434,703	1.6 %
cc-news	313,374	—	195,910,044	1.1 %
ultrachat	71,981	—	92,174,421	0.5 %
tiny-textbooks	130,174	—	85,612,490	0.5 %
Region	14,670,308	1,480,815	18,266,295,978	100 %

Table 5: The stage-2 region by source, after the scans (packing manifest; units are documents or ARC copies, each ending with an end-of-text token). The prefix of the region with the length of the 2-billion-token ablation data has a pool share of 11.70 % versus 11.79 % in the whole region.

5 Contamination control and held-out sets

5.1 Scans of the top-up

Scan	Rule	Rows cut
WikiText-2 and ARC-Easy/Challenge (test, validation)	any shared normalized 13-gram; ARC questions of 6–12 words matched exactly	4,367
Held-out control sets (Section 5.2)	≥ 10 distinct 13-grams shared with one control document	21,765
BLiMP	sentences of ≥ 6 words	0
Union	a cut row is removed with all its ARC copies	25,808 (0.23 %)

Table 6: Scans before packing. The largest cuts: fwe-expansion 12,521, dclm-baseline 5,711, fineweb-edu 4,962, wikipedia 1,965.

5.2 Held-out evaluation axes

- **DCLM axis:** 3,457 documents from `allenai/DataDecide-data-recipes@3baf34ba`, recipe `v0_fw_top3p`, part `part-00-00000` (SHA `ff7cecf4`); 1,543 of 5,000 dropped for overlap with any training slice or board tests.
- **FineWeb-Edu axis:** 3,507 documents from `sample/100BT/000_00009` (SHA `1b9f68e5`); 1,493 of 5,000 dropped for 13-gram or URL overlap.

Both axes are measured in bits per byte, so different tokenizers are compared fairly.

6 Considered and rejected

Option	Why not	Evidence
Repeat the pool (~ 2.65 epochs)	lost the ablation	64M ablation (b)
100 % FineWeb-Edu	better on its own domain only: FWE -0.016 bpb, DCLM $+0.089$ bpb	64M ablation (e)
FineWeb-Edu-heavy decay	DCLM $+0.0505$ bpb, board score $+0.07$ (noise)	H2, four runs
12k tokenizer	$+0.0045$ bpb at equal parameter count	ablation (h)
32k tokenizer	not better than 24k ($+0.0014$ / $+0.0017$ bpb)	ablation (h)
SmolLM2 49k tokenizer	not yet compared ; planned as ablation (t)	—
QC-FineWeb top 3 %	same distribution as the DCLM control axis; licence unclear	decision of 6 October
arXiv	rejected in an earlier corpus review (25 September)	—
PubMed beyond PMC commercial	licence per article	PMC OA commercial used
OpenSubtitles	only a 6.7 MB English file available	dropped
One StackExchange URL range	Windows Defender flagged Trojan:HTML/ScrInject	excluded by name
CC BY-NC-SA documents	non-commercial licence	removed (1,175)

Table 7: Options we considered and rejected. Ablations are 64M-parameter models trained on 2 billion tokens; bpb is measured on the held-out axes of Section 5.2 (negative is better).

7 Tokenizers

Tokenizer	SHA	Notes
12k (v5)	37333075	vocabulary 12,288
24k (Slayer 150 1.01)	afeb22b4	byte-level BPE, vocabulary 24,576; trained on 1.0 GB of English sampled from the pool (205,522 documents); 4.021 bytes per token on held-out English, no round-trip failures
32k	30be1bba	same trainer and sample; 4.126 bytes per token

Table 8: Tokenizers.

8 Reproducing

Every artefact is pinned by SHA-256. The pool artefacts (pool 12k, pool 24k, stage-1 queue) were rebuilt bit for bit on at least two machines and the raw file was verified after each download. The top-up build was made once and is checked file by file against its manifest. Tokenization of

the top-up was checked identical on two machines (the same file gives the same token stream SHA).

Artefact	SHA-256 (prefix)
ARC-MIX raw file (12k)	e87f594f
ARC-MIX pool (12k)	ecfd0a40
Pool, 24k	8911b9c5
Stage-1 queue (8 shards $\times$ 32,835 steps)	c60d76c9
Top-up fetch plan / fetch manifest / sources	5817b464 / 6a480f54 / 38489e4e
Top-up build manifest	b73f45f5
Scan cut lists (packing arguments)	baa6a4af
Packing code	3976d146

Table 9: Artefacts and their SHA-256 prefixes.

9 Open items

- Licences to confirm before the model card: open-web-math, StackExchange (content CC BY-SA), Project Gutenberg (public domain in the US), cc-news (unknown, named explicitly).
- The data-mixture test (g) on a 2-billion-token prefix of the region (SHA cf67e423) against the 24k control, with its non-inferiority rule fixed before the run.